Smith Mbele
ST10472346

Corrections: Endpoint Plan and SQL Script

<table>
<tr><th>Lecturer comment</th><th>Current submission</th><th>What I fixed</th></tr>
<tr><td>Routes not under /api/</td><td>Paths are like /auth/users, /events, /categories.</td><td>
1. Authentication and Users
<table>
<tr><th>Operation</th><th>Method</th><th>Corrected endpoint</th></tr>
<tr><td>User Registration</td><td>POST</td><td><code>/api/auth/users</code></td></tr>
<tr><td>User Login</td><td>POST</td><td><code>/api/auth/login</code></td></tr>
<tr><td>Get User Profile</td><td>GET</td><td><code>/api/profiles/{user_id}</code></td></tr>
<tr><td>Update Organiser Profile</td><td>PUT</td><td><code>/api/organisers/{organiser_id}</code></td></tr>
<tr><td>Update Participant Profile</td><td>PUT</td><td><code>/api/participants/{participant_id}</code></td></tr>
<tr><td>Delete User Account</td><td>DELETE</td><td><code>/api/users/{user_id}</code></td></tr>
</table>
2. Events
<table>
<tr><th>Operation</th><th>Method</th><th>Corrected endpoint</th></tr>
<tr><td>Create Event</td><td>POST</td><td><code>/api/events</code></td></tr>
<tr><td>List All Events</td><td>GET</td><td><code>/api/events</code></td></tr>
<tr><td>Get Event Details</td><td>GET</td><td><code>/api/events/{event_id}</code></td></tr>
<tr><td>Update Event</td><td>PUT</td><td><code>/api/events/{event_id}</code></td></tr>
<tr><td>Delete Event</td><td>DELETE</td><td><code>/api/events/{event_id}</code></td></tr>
</table>
3. Categories
<table>
<tr><th>Operation</th><th>Method</th><th>Corrected endpoint</th></tr>
<tr><td>Create Category</td><td>POST</td><td><code>/api/categories</code></td></tr>
<tr><td>List All Categories</td><td>GET</td><td><code>/api/categories</code></td></tr>
<tr><td>Get Category Details</td><td>GET</td><td><code>/api/categories/{category_id}</code></td></tr>
<tr><td>Update Category</td><td>PUT</td><td><code>/api/categories/{category_id}</code></td></tr>
<tr><td>Delete Category</td><td>DELETE</td><td>/api/category/{category_id}</td></tr>
</table>
</td></tr>
</table>

4. Enrolments

Operation	Method	Corrected endpoint
Enrol Participant in Event	POST	/api/enrolments
List Event Enrolments	GET	/api/enrolments
Get Enrolment Details	GET	/api/enrolments/{enrolment_id}
Update Enrolment Status	PUT	/api/events/{event_id}/enrolments/{enrolment_id}
Participant Withdraws from Event	DELETE	/api/events/{event_id}/enrolments/{enrolment_id}

5. Routes

Operation	Method	Corrected endpoint
Create Route for an Event	POST	/api/events/{event_id}/routes
List Routes for the Event	GET	/api/events/{event_id}/routes
Get Route Details	GET	/api/events/{event_id}/routes/{route_id}
Update Route	PUT	/api/events/{event_id}/routes/{route_id}
Delete Route	DELETE	/api/events/{event_id}/routes/{route_id}

6. Results

Operation	Method	Corrected endpoint
Record Race Results	POST	/api/events/{event_id}/results
Race Results	GET	/api/results/{results_id}
Performance History	GET	/api/results/history/{participant_id}
Update Race Results	PUT	/api/results/{results_id}
Delete Race Results	DELETE	/api/results/{results_id}

Failure responses limited

Added the failure responses to the endpoints:

```
{
  "error": "Invalid request",
  "message": "The supplied data is invalid or incomplete."
}
400 – Bad Request
```

```
{
  "error": "Unauthorized",
  "message": "Authentication is required or the credentials are invalid."
}
401 – Unauthorized

{
  "error": "Forbidden",
  "message": "You do not have permission to perform this action."
}
403 – Forbidden

{
  "error": "Not found",
  "message": "The requested resource does not exist."
}
404 – Not Found
```

I will apply those failure responses to the following endpoints:

Registration	400, 409
Login	401
Get /api/update/delete profiles	401, 403, 404
Create /api /update/delete events	400, 401, 403, 404
Create /api /update/delete categories	400, 401, 403, 404
Get /api/list categories	404
Enrolments	400, 401, 403, 404
Routes	400, 401, 403, 404
Results	400, 401, 403, 404

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<table>
<tr>
<td>Token auth instead of session auth</td>
<td>Response:
/api/auth/login

<pre>{
"user_id": "...",
"username": "...",
"email": "...",
"role": "...",
"token": "..."
}</pre></td>
<td>Response:
/api/auth/login

<pre>{
  "user_id": "...",
  "username": "...",
  "email": "...",
  "role": "..."
  "message": "Login successful"
}</pre></td>
</tr>
<tr>
<td>The required Role Required column is missing</td>
<td></td>
<td>
<table>
<tr><th>Area</th><th>Operation</th><th>HTTP Method</th><th>Role Required</th></tr>
<tr><td>Auth</td><td>User Registration</td><td>POST</td><td>Public</td></tr>
<tr><td>Auth</td><td>User Login</td><td>POST</td><td>Public</td></tr>
<tr><td>Profile</td><td>Get own profile</td><td>GET</td><td>Logged in</td></tr>
<tr><td>Profile</td><td>Update organiser profile</td><td>PUT</td><td>Organiser</td></tr>
<tr><td>Profile</td><td>Update participant profile</td><td>PUT</td><td>Participant</td></tr>
<tr><td>Profile</td><td>Delete own account</td><td>DELETE</td><td>Logged in</td></tr>
<tr><td>Events</td><td>Create event</td><td>POST</td><td>Organiser</td></tr>
<tr><td>Events</td><td>List all events</td><td>GET</td><td>Logged in</td></tr>
<tr><td>Events</td><td>Get event details</td><td>GET</td><td>Logged in</td></tr>
<tr><td>Events</td><td>Update event</td><td>PUT</td><td>Organiser</td></tr>
<tr><td>Events</td><td>Delete event</td><td>DELETE</td><td>Organiser</td></tr>
<tr><td>Categories</td><td>Create category for event</td><td>POST</td><td>Organiser</td></tr>
<tr><td>Categories</td><td>List categories for event</td><td>GET</td><td>Logged in</td></tr>
<tr><td>Categories</td><td>Get category details</td><td>GET</td><td>Logged in</td></tr>
<tr><td>Categories</td><td>Update category</td><td>PUT</td><td>Organiser</td></tr>
<tr><td>Categories</td><td>Delete category</td><td>DELETE</td><td>Organiser</td></tr>
<tr><td>Enrolments</td><td>Enrol in event with a category</td><td>POST</td><td>Participant</td></tr>
<tr><td>Enrolments</td><td>List my enrolments</td><td>GET</td><td>Participant</td></tr>
<tr><td>Enrolments</td><td>List enrolments for an event</td><td>GET</td><td>Organiser</td></tr>
<tr><td>Enrolments</td><td>Get enrolment details</td><td>GET</td><td>Participant or Organiser</td></tr>
<tr><td>Enrolments</td><td>Update enrolment status</td><td>PUT</td><td>Organiser</td></tr>
<tr><td>Enrolments</td><td>Withdraw from event</td><td>DELETE</td><td>Participant</td></tr>
<tr><td>Routes</td><td>Create route</td><td>POST</td><td>Organiser</td></tr>
</table>
</td>
</tr>
</table>

			Routes	List routes for event	GET	Logged in
			Routes	Get route details	GET	Logged in
			Routes	Update route	PUT	Organiser
			Routes	Delete route	DELETE	Organiser
			Results	Record race result for an enrolment	POST	Organiser
			Results	Get result details	GET	Participant or Organiser
			Results	Get participant performance history	GET	Participant
The script still requires correction to run cleanly from a fresh SQL Server environment.		``` USE master; GO IF DB_ID(N'RaceDay') IS NOT NULL BEGIN ALTER DATABASE RaceDay SET SINGLE_USER WITH ROLLBACK IMMEDIATE; DROP DATABASE RaceDay; END GO CREATE DATABASE RaceDay; GO USE RaceDay; GO ```
Enrolment sample data is incomplete		``` INSERT INTO ENROLLMENT (participant_id, event_id, category_id, status) VALUES (1, 1, 3, 'confirmed'), -- Thabo Molefe, Johannesburg City Run, 10km Open (2, 1, 2, 'registered'), -- Lindiwe Dlamini, Johannesburg City Run, 5km Women (1, 2, 4, 'registered'), -- Thabo Molefe, Pretoria Spring Marathon, 10km Men (2, 3, 7, 'confirmed'); -- Lindiwe Dlamini, Soweto Heritage Trail, 5km Trail ```

Events are missing Distance and EventType		CREATE TABLE EVENT (event_id INT PRIMARY KEY IDENTITY(1,1), event_name VARCHAR(200) NOT NULL, description NVARCHAR(MAX) NULL, location VARCHAR(200) NOT NULL, latitude DECIMAL(10, 8) NOT NULL, longitude DECIMAL(11, 8) NOT NULL, event_date DATETIME2 NOT NULL, distance DECIMAL(6, 2) NOT NULL, event_type VARCHAR(10) NOT NULL CHECK (event_type IN ('run', 'walk', 'cycle')), status VARCHAR(50) NOT NULL CHECK (status IN ('scheduled', 'ongoing', 'completed', 'cancelled')), created_at DATETIME2 DEFAULT GETUTCDATE(), updated_at DATETIME2 NULL); INSERT INTO EVENT (event_name, description, location, latitude, longitude, event_date, distance, event_type, status) VALUES ('Johannesburg City Run', 'A city centre road race through Sandton.', 'Sandton, Johannesburg', -26.1076, 28.0567, '2026-10-10 07:00:00', 10.00, 'run', 'scheduled'), ('Pretoria Spring Marathon', 'A spring road race through the Pretoria CBD.', 'Pretoria CBD', -25.7479, 28.2293, '2026-10-24 06:30:00', 42.20, 'run', 'scheduled'), ('Soweto Heritage Trail', 'A guided walking trail through Soweto.', 'Soweto, Johannesburg', -26.2485, 27.8540, '2026-11-07 07:30:00', 10.00, 'walk', 'scheduled');
ERD and SQL inconsistencies for Events, Categories, and Enrolments	Categories	CREATE TABLE CATEGORY (category_id INT PRIMARY KEY IDENTITY(1,1), event_id INT NOT NULL, category_name VARCHAR(100) NOT NULL, created_at DATETIME2 DEFAULT GETUTCDATE(), FOREIGN KEY (event_id) REFERENCES EVENT(event_id) ON DELETE CASCADE, UNIQUE (event_id, category_name));

		INSERT INTO CATEGORY (event_id, category_name) VALUES (1, '5km Men'), (1, '5km Women'), (1, '10km Open'), (2, '10km Men'), (2, '10km Women'), (2, '21km Open'), (3, '5km Trail'), (3, '10km Trail');
	Enrolments	CREATE TABLE ENROLLMENT (enrolment_id INT PRIMARY KEY IDENTITY(1,1), participant_id INT NOT NULL, event_id INT NOT NULL, category_id INT NOT NULL, results_id INT UNIQUE, entry_date DATETIME2 DEFAULT GETUTCDATE(), status NVARCHAR(50) NOT NULL CHECK (status IN ('registered', 'confirmed', 'started', 'finished', 'withdrawn', 'disqualified')), FOREIGN KEY (participant_id) REFERENCES PARTICIPANT(participant_id) ON DELETE CASCADE, FOREIGN KEY (event_id) REFERENCES EVENT(event_id) ON DELETE CASCADE, FOREIGN KEY (category_id) REFERENCES CATEGORY(category_id), FOREIGN KEY (results_id) REFERENCES RESULTS(results_id) ON DELETE SET NULL, UNIQUE (participant_id, event_id));